

Abhishek Singh

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Education

Indian Institute of Technology Patna 2023–2027 (expected)
B.Tech in Electronics and Communication Engineering; Semester VII CPI: 7.00/10
Completed ML / computing coursework: Introduction to AI–ML; Programming and Data Structures with Laboratory; Random Signals and Stochastic Processes; Digital Signal Processing; Signals and Systems.
Completed embedded / hardware coursework: Microcontroller and Embedded System; Microprocessor; Internet of Things; Computer Organization and Architecture; Digital Electronics; VLSI Design; Control Systems.

Technical Skills

Languages: Python, C++, C, SQL
Machine Learning: PyTorch, TensorFlow, CNNs, autoencoders, computer vision, anomaly detection
Data / Engineering: NumPy, Pandas, OpenCV, Git, Linux, AWS EC2
Embedded / Simulation: ESP32, Internet of Things, MEEP (FDTD)

Research Experience

Photonic Neural Computing Research — IIT Patna May 2026–Present
Under Prof. Sumanta Gupta, Photonics / Optical Communication Group

- Investigating photonic–electronic neural-network primitives, focusing on optical multiply–accumulate (MAC) operations for silicon photonic hardware.
- Studying microring-resonator weighting and balanced detection for signed-weight computation toward a photoelectronic multiply–accumulate neuron (PEMAN) architecture.

Video Surveillance Research Intern — IIT Patna May–July 2026
Under Prof. Mahesh Kumar Kolekar Project: AI Surveillance System

- Developed a PyTorch autoencoder framework for video anomaly detection, including training, inference, and evaluation pipelines.
- Designed Hybrid Patch Feature Scoring to combine patch-level reconstruction information with feature-based anomaly scoring.
- Trained and evaluated models on UCSD Pedestrian, CUHK Avenue, and ShanghaiTech surveillance datasets.

Independent Research — Silicon Photonic Microring Resonators May–June 2026

- Simulated lateral, vertical/3D, and parallel-cascaded coupling configurations in MEEP; analyzed coupling-gap sweeps, critical coupling, quality factor, and numerical convergence.

Engineering Experience & Projects

Software Head — Inditronix AI Labs April 2025–May 2026

- Led software architecture and development for healthcare AI solutions and coordinated a multidisciplinary software team.
- Deployed infrastructure on AWS EC2 and developed ESP32-based IoT applications.

USAS Ingestor Engine Private repository

- Built a multi-camera CCTV and DOT stream ingestion pipeline with synchronization, preprocessing, and storage modules for surveillance analytics.

Kaggle Machine Learning Competitions

- Participated in healthcare, finance, and environmental prediction competitions: Problematic Internet Use, Loan Payback, and Rainfall Prediction.

Achievements & Competitive Programming

- GATE ECE 2026:** All India Rank 1046; score 662. **JEE Advanced:** AIR 5701; **JEE Main:** AIR 4311.
- Codeforces:** Pupil, peak rating 1241; best Div. 2 rank 2295. **CodeChef:** 3-star, peak rating 1643; best ranks 150 (Div. 3) and 459 (Div. 2).

Affiliation: IEEE Student Member, IEEE Photonics Society (since June 2026).